Free fall

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe and calculate ideal free fall near Earth.

SUCCESS CRITERIA - I can...

1. State the no-air-resistance assumption.
2. Use equal velocity changes in equal times.
3. Separate acceleration from mass and current velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “free fall” and “initial” to explain your model or result. Sentence starters are optional.

free fall: motion with gravity as the only force

initial: at the beginning of the interval; subscript i

acceleration: rate of change of velocity

01 Bring back an earlier idea

A toy model predicts position (-40, 60) km with bounds ±4 km in x and ±6 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_f = v_i + g\Delta t$; $\Delta y = v_i\,\Delta t + \frac{1}{2}g(\Delta t)^2$.
Downward positive for these equations; $g = 10$ m/s². Ideal free fall: gravity is the only force.

Worked reasoning: A ball dropped from rest has downward speeds 0, 10, 20 m/s at 0, 1, 2 s in ideal free fall with g=10 m/s². Equal times give equal velocity changes.

Your turn: Sketch its velocity-time graph with downward positive.
Explain why the positions are not equally spaced.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

03 Represent and interpret

Two balls of different mass are released together in a vacuum from the same height.
Compare acceleration and arrival time.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Solve and justify

A ball is thrown upward at 20 m/s.
Use upward positive.
Find time to the top and velocity at 3 s.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

At the highest point of a tossed ball, a classmate says gravity stops because speed is zero. Correct the claim.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

An a-versus-F_{net} graph has slope 2 (m/s²)/N.
Find mass and acceleration at 3 N.
Explain why a driving-force plot can have an offset.

Word help: mass = a measure of how strongly an object resists acceleration

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

07 Exit check

A ball starts with downward velocity 5 m/s and falls for 2 s.
Find final speed and displacement.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A ball is thrown up at 15 m/s from a 20 m platform.
Use g=10 m/s².
Find its height at 2 s and its time to reach the ground.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3